



Department of Education

**STEM Education
National Schools of Excellence
Teacher's Guide
Grade 12
Term 3**

<u>DOMAIN</u>	<u>Phylum</u>
SCIENCE	Chemistry
Technology	
Engineering	
Mathematics	

Issued free to schools by the Department of Education

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SECRETARY’S MESSAGE

This Chemistry Syllabus is to be used exclusively by teachers of Chemistry to teach Grade 11 Students enrolling for the STEM Education through the Schools of Excellence throughout Papua New Guinea. The STEM Chemistry Bridging Syllabus ensured that Grade 11 students enrolling for STEM Education were grounded in the basic concepts espoused in Chemistry before undertaking the advanced concepts introduced in this syllabus.

The syllabus conforms to the visions of the National Education Plan and the School of Excellence Policy. Furthermore, it is undergirded by a policy and legislative matrix that were issued by the National Government over the last decades, such as the Vision 2050 and Development Strategic Plan 2030. The Preamble of the National Constitution, especially the First Directive of the *National Goals and Directive Principles* makes a clarion call for the Integral Human Development, which the Department took heed of to ensure, through this new syllabus, that *all forms of beneficial creativity, including sciences and cultures, [are] to be actively encouraged*. A principle objective of the *National Education Act 1983 (amended 1995)* reiterates this virtue of Integral Human Development (Section 4).

In accordance with the widespread demand for a home-grown curriculum to replace the existing one with the view to position the country as a bastion of innovation in the new century the Secretary of Education has embarked on introducing the STEM Curriculum to the Schools of Excellence. It is hoped that students coming through the STEM Education will not only gain productive employment in the knowledge-based economy but become creators, innovators and inventors of knowledge and ultimately commercialize their creativity through development of their intellectual property rights.

In addition, there are four core elements espoused in this syllabus, insofar as student development is concerned: Students are expected to gain knowledge and skills and apply them to solve every day problems whilst appreciating Chemistry as a fundamental field of science.

This STEM Chemistry Syllabus for Grade 11 constitutes ten (10) Thematic Area (Thema) and is intended for students enrolling for STEM Education. In this booklet only three (3) Thema are presented. By studying these Thema students are expected to gain the foundational knowledge needed in Chemistry.

The success of this new Curriculum is highly dependent on the Teachers. They play a pivotal role by being innovative and keeping abreast of new information on scientific research and innovations. The challenge for teachers of Chemistry is to engage students to learn realistic context for increased and better understanding and appreciate the relevance of Chemistry to human lives, including good health and environment.

In order to ensure Teachers discharge their responsibilities adequately this syllabus is accompanied by a Teacher’s Guide and Teacher’s Resource Book which provides sufficient teaching and assessment materials to help Teachers.

I commend and approve this syllabus as the official curriculum for Chemistry to be used for the Grade 11 STEM Education in the Schools of Excellence throughout Papua New Guinea.

DR UKE KOMBRA, PHD
Secretary for Education

INTRODUCTION

Chemistry is the study of matter and its composition, how they interact to form new compounds through chemical reactions and the energy they generate as a result of these interactions. The concepts taught in Chemistry help us to understand these interactions at the molecular and atomic levels. Every substance, whether naturally occurring or artificially produced, consists of one or more elements of the periodic table.

Since Chemistry is an experimental science, laboratory work becomes an essential part of the syllabus. Practical work can be used to enable students to investigate the properties and reactions of substances and to provide opportunities for students to learn and test the principles and concepts of Chemistry.

The Grade 11 Chemistry Syllabus entails ten (10) Thematic Areas (Thema) and these are listed below. Each Thema has a single Module and three topics. Over time different Modules will be introduced.

Since Chemistry is a highly specialized subject students undertaking Chemistry are expected to possess high levels of cognitive and numeracy competency, and a firm grasp of the language skills.

The modules in this Syllabus are sequenced to guide teachers of Chemistry through the process of imparting basic chemistry knowledge and skills to students.

THEMA	MODULE	TOPICS
BIOTECHNOLOGY	12.1 Protein Sequencing	1. Amino Acids 2. Protein Structure and Function 3. Protein Purification 4. Protein Sequencing
HEALTH AND MEDICINE	12.1 Drug Discovery, Design and Development	1. Drug Targets 2. Drug Discovery 3. Computer Aided Drug Design 4. Drug Development and Production
ENERGY	12.1 Hydrogen – Future of Energy	1. Hydrogen as Fuel 2. Methods of Producing Hydrogen 3. Storing Hydrogen 4. Hydrogen Fuel Cell
BATTERY TECHNOLOGY	12.1 Solid-State Battery	1. Introduction to Solid-State Battery 2. Solid Electrolyte 3. Manufacturing process Solid-State Battery
ARTIFICIAL INTELLIGENCE	12.1 Artificial Molecular Machines	1. Introduction to Artificial molecular Machines 2. Catenanes and Rotaxanes 3. Other Protein based Motors under development
ROBOTICS	12.1 Robotics Meets Chemistry	1. Acidity Theories 2. Robotics 3. Programming a Robot

RATIONALE

There is a confluence of events that necessitated the need to introduce STEM Curriculum to the Schools of Excellence:

1. The School of Excellence Policy that was launched by Prime Minister Honorable James Marape, MP, in June 2020, calls for the establishment of the Schools of Excellence and the introduction of STEM Curriculum;
2. The Government has decided to replace the Observation Based Education to Standards Based Education in 2019 and tasked the Department of Education to introduce the STEM Curriculum as part of its reform agenda;
3. There is public outcry, both from public and private sectors, over the quality of graduates coming through the National Education System; and
4. The changing landscape in the technology domain globally and the pressing need to produce the next generation of workforce that is agile and can adopt to the changing circumstances as well as become enterprising individuals through their creation of intellectual properties.

The combination of these factors has led the Department of Education to adopt the STEM Curriculum and to introduce this to the Schools of Excellence as a necessary disruptive intervention in an effort to develop the next generation of national intellectual capital and to drive the growth and prosperity of the country through creation of intellectual properties.

In developing the STEM Curriculum, it was realized that the Grades 9 and 10 Science Syllabi does not provide a firm foundation to launch students into STEM Curriculum at the Schools of Excellence. Consequently, a STEM Bridging Program was necessary. The STEM Bridging Syllabus is a bridge across the divide that Grade 11 Students enrolling for STEM Education must cross.

In the Chemistry Bridging Syllabus students were introduced to the basic concepts and knowledge of Chemistry in order for them to develop an appreciation of the discipline in a broader context. In this syllabus students are introduced to advanced concepts in Chemistry as they are applied to the ten Thema.

The introduction of the STEM Curriculum, especially the Bridging Program and the Grades 11 and 12 Syllabi, will cause consequential changes to the Upper Classes of STEM Education and the Lower Secondary Syllabi. These syllabi will be introduced in the course of time.

STEM DOMAIN

At this juncture of our national development there is no other task before us more important than deploying a robust STEM Curriculum that develops the next generation of highly educated students to enter the work force. By building on the best of current practices and standards the STEM Curriculum aims to take us, beyond the constraints of the existing structures of schooling, towards a shared vision of excellence nationally and marketing the nation globally in terms of Science, Technology, Engineering and Mathematics. In this STEM Curriculum, there are ten (10) Thema (Thematic Areas) introduced.

Science

It is the study of the natural world, including the laws of the nature associated with physics, chemistry and biology and the treatment or application of facts, principles, concepts or conventions associated with these disciplines. Science is both a body of knowledge that has been accumulated over time and a process-scientific inquiry-that generates new knowledge. Knowing from science informs the engineering design process. In this STEM Curriculum the Science Domain has three Phyla – Biology, Chemistry and Physics.

Technology

Comprises the entire system of people and organizations, knowledge, processes, and devices that go into creating and operating technological artifacts, as well as the artifacts themselves. Throughout history humans have created technology to satisfy their wants and needs. Much of modern technology is the product of science and engineering and technological tools used in both fields. There are two Phyla – Information Technology and Computer Sciences – introduced in this STEM Curriculum for the Technology Domain.

Engineering

Concerns both the knowledge about the design and creation of products and a process for solving problems. The design process must result in a final product that must operate or function under various constraints. One constraint in engineering design is the laws of nature. Engineers must learn to harness the natural forces at work around them and through their design process develop products that must operate in harmony with nature. Other constraints include things such as time, money, available materials, ergonomics, environmental regulations and manufacturability. Engineering utilizes concepts in science and mathematics as well as technological tools. There are three (3) Phyla introduced in the Engineering Domain – Civil, Mechanical and Electrical.

Mathematics

It is the study of patterns and relationships among quantities, numbers, and shapes. Specific branches of mathematics include arithmetic, geometry, algebra, trigonometry and calculus. Mathematics is used in science and in engineering. In this STEM Curriculum, Mathematics is considered as a Service Domain; specific topics tailored towards specific Thema are also considered relevant to undergird the relevance and applicability of Mathematics Domain to other fields of knowledge.

STEM CHEMISTRY CONTENT OVERVIEW – GRADE 12, TERM 3

Table 1.1 Sequencing & Benchmarking STEM Chemistry Content Standards

Setting STEM Chemistry Content Standards for Progressive Learning & Assessment			
Thema	Artificial Intelligence	Module Content Standard	
Module 12.1 Artificial Molecular Machines		12.1 To familiarize oneself with the potential of creating molecular machines.	
Topics	Performance Indicators	Module Benchmark	Assessment Tasks
1. Introduction to Artificial Molecular Machines	12.1.1a: Define Artificial Molecular Machines 12.1.1b: Discuss certain chemical molecular machines that has been identified.	Students' ability to master the knowledge and skills of: • Artificial Molecular Machines • Catenanes • Rotaxanes	Assignment + Test + Quiz
2. Catenanes and Rotaxanes	12.1.2a: Explain and show with a diagram Catenanes are considered as Artificial Molecular Machines 12.1.2b: Explain and show with a diagram how Rotaxanes are considered as Artificial Molecular Machines		
3. Other Protein based motors under Development	12.1.3a: Describe and explain with an example the different Artificial molecular machines that have been discovered.		

Thema	Robotics	Module Content Standard	
Module 12.1 Robotics Meets Chemistry		12.1: Design and build a Chembot that tells whether or not the pH of a liquid as changed with a change in temperature.	
Topics	Performance Indicators	Module Benchmark	Assessment Tasks
1. Acidity Theories (Recall)	12.1: Design and build a Chembot that tells whether or not the pH of a liquid as changed with a change in temperature.	Students master the knowledge and skills of: - Assembling a robot - Programming a robot to solve a problem or complete a task	Practical Assignment + Test
2. Robotics (Basics)	12.1.1a: Explain what Bronsted-Lowry Theory, Arrhenius Theory and Lewis Acid-Base theory are and how they are related 12.1.1b: Define what pH is 12.1.1c: Explain Temperature and pH relationship		
3. Programming a Robot	12.1.3a: Explain the different programming languages used in robotics		

Table 1.2 Mapping Content Progressions by Modules

Thema/Module	Module Content Standard	12.1	12.1
ARTIFICIAL INTELLIGENCE / 12.1 Artificial Molecular Machines	12.1 To familiarize oneself with the potential of creating molecular machines.	✓	
ROBOTICS / 12.1 Robotics Meets Chemistry	12.1: Design and build a Chembot that tells whether or not the pH of a liquid as changed with a change in temperature.		✓

THEMA: ARTIFICIAL INTELLIGENCE

Introduction

In today's world, technology advances at a breakneck pace, and we are constantly exposed to new innovations. Artificial Intelligence is one of the burgeoning computer science technologies that is poised to usher in a new era in the world by creating intelligent machines. Artificial Intelligence is currently everywhere. It is now working on a wide range of subfields, from broad to specialized, such as self-driving vehicles, chess, proving theorems, music, painting, and so on.

AI is one of the most intriguing and ubiquitous areas of computer science, with a bright future ahead of it. AI could make a machine function like a person. It is made up of the terms Artificial and Intelligence, with Artificial referring to "man-made" and Intelligence referring to "thinking power," therefore AI refers to "a man-made thinking power." As a result, we may define AI as is a field of computer science that allows us to construct intelligent robots that can behave, think, and make decisions like people.

Artificial intelligence arises when a machine can learn, reason, and solve problems in the same way that humans do. You don't need to pre-program a machine to accomplish some job using Artificial Intelligence; but you may construct a computer with programmed algorithms that can work with its own intelligence, which is AI's magnificence

Artificial Intelligence is not only a branch of computer science, even though it is so broad and encompasses a wide range of other elements. To build AI, we must first understand how intelligence is made up. Intelligence is an intangible component of our brain that is made up of reasoning, learning, problem-solving vision, language comprehension, and so on.

Artificial Intelligence requires the following discipline to attain the above criteria for a machine or software:

- Mathematics
- Biology
- Psychology
- Sociology
- Computer Science
- Neurons Study
- Statistics

MODULE 12.1: ARTIFICIAL MOLECULAR MACHINES

4-5 Weeks

Machines have revolutionized civilization and improved our quality of life. What's next once we've moved from mass-producing extremely large computer machine technology to sending a man to the moon? Many chemists are now looking at the idea of constructing gadgets from molecules. Because they are man-made or "artificial," these molecules are known as Artificial Molecular Machines.

Knowing how to control motion on a miniscule scale necessitates some chemical knowledge, yet tiny machines have the potential to revolutionize everything from health to material science, where molecular processes are crucial. A machine is any device that turns energy into at least one moving part – each with its own function – and then combines these parts to produce a useful motion as an output known as "work."

Consider an old watch: all of the interconnecting cogs are arranged in such a manner that the hands on the dial move at the correct rate.

Making machines smaller has several obvious advantages, including the ability to transport them more easily and move them more precisely.

Linking Modules

Prerequisites: NIL

Broad Content Standard

12.1 To familiarize oneself with the potential of creating molecular machines.

Content Benchmark

Students' ability to master the knowledge and skills of:

- Artificial Molecular Machines
- Catenanes
- Rotaxanes

Topic 1: Introduction to Artificial Molecular Machines

Performance Indicators

Students are expected to:

12.1.1a: Define Artificial Molecular Machines

12.1.1b: Discuss certain chemical molecular machines that has been identified.

No Activity

Topic 2: Catenanes and Rotaxanes

Performance Indicators

Students are expected to:

12.1.2a: Explain and show with a diagram Catenanes are considered as Artificial Molecular Machines

12.1.2b: Explain and show with a diagram how Rotaxanes are considered as Artificial Molecular Machines

No Activity

Topic 3: Other Protein based Motors Under Development

Performance Indicators

Students are expected to:

12.1.3a: Describe and explain with an example the different Artificial molecular machines that have been discovered.

Activity 1: Group Presentation

Research a particular type of Molecular Machine component that has been discovered and explain how it has a signification application.

SAMPLE ASSESSMENT

1. Sample Assessment Methods

- Group Assignment
- Presentation
- Projects
- Written or Practical Quiz
- Written or Practical Test
- Practical Assignment
- Case Studies

2. Sample Assessment Tasks

Module	Content Standard	Topic	Assessment Tasks	
12.1: Artificial Molecular Machines	12.1 To familiarize oneself with the potential of creating molecular machines.	1. Introduction to Artificial Molecular Machines	Research Assignment	Written Test
		2. Catenanes and Rotaxanes		
		3. Other Protein based under Development		

3. Sample Assessment Marking Guide

Topic	Assessment Task	Assessment Criteria	Total Marks
Subtotal			20
Module 12.1 (All four Topics)	Research Assignment	Research a particular type of Molecular Machine component that has been discovered and explain how it has a signification application.	20
			20
			10
Subtotal			50
Module 12.1 (All four Topics)	Written Test (Multiple Choices, Short Answers, Calculations, etc)	Correctly answers all long questions	10
		Correctly answers all multiple choice	15
		Correctly answers short answer questions	5
		Subtotal	30
Module TOTAL			100

THEMA: ROBOTICS

Introduction

Robotics is a multidisciplinary discipline combining computer science and engineering. It is the study of the design, building, operation, and use of robots. Robotics aims to create devices that can aid and benefit people.

Mechanical engineering, electrical engineering, information engineering, mechatronics, electronics, bioengineering, computer engineering, control engineering, software engineering, and mathematics are just a few of the areas that robotics encompasses.

Robotics is the study of creating machines that can perform the function of people and mimic their behaviors. Robots can be employed in a variety of settings and for a variety of reasons, but many are now utilized in hazardous areas (such as radioactive material inspection, bomb detection and deactivation), manufacturing operations, or in conditions where humans are unable to live (e.g. in space, underwater, in high heat, and clean up and containment of hazardous materials and radiation).

MODULE 12.1: ROBOTICS MEET CHEMISTRY

1 Week

This module is more project-based and is designed to simply have students design a Chembot – a robot that does simple chemistry experiments. The aim is to have students see the link between branches of science – in this case, chemistry with robotics, mechanical engineering, electrical engineering, and programming. The challenge presented in this module is for students to build a robot to answer the question: “When temperature changes, does it affect the pH of a liquid or not?”

Linking Modules

Prerequisites: NIL

Broad Content Standards

12.1: Design and build a Chembot that tells whether or not the pH of a liquid as changed with a change in temperature.

Content Benchmarking

Students master the knowledge and skills of:

- Assembling a robot
- Programming a robot to solve a problem or complete a task

Topic 1: Acidity Theories (Recall)

Performance Indicators

Students are expected to:

- 12.1.1a:** Explain what Bronsted-Lowry Theory, Arrhenius Theory and Lewis Acid-Base theory are and how they are related
- 12.1.1b:** Define what pH is
- 12.1.1c:** Explain Temperature and pH relationship

No Activity

Topic 2: Robotics (Basic)

Performance Indicators

Students are expected to:

- 12.1.2a:** Explain the components of robots

No Activity

Topic 3: Programming a Robot

Performance Indicators

Students are expected to:

12.1.3a: Explain the different programming languages used in robotics

Activity Project

Title: Building a Chembot

Setting up the challenge

This project is all about collaborative thinking and working. Students use Lego Mindstorms robotics kits to create the robots, which they then outfit with sensors to assess the temperature and pH of a liquid. You may utilize a three-step, progressively collaborative process to assist students create and enhance their first thoughts about how to address the challenge:

1. *Think:* Students think independently about how to proceed, forming ideas of their own.
2. *Pair:* Students discuss their thoughts in pairs. This step allows students to articulate their ideas and to consider those of another person.
3. *Share:* Student groups share their ideas with a larger group, such as the whole class. Often students are more comfortable presenting ideas to a group with the support of a partner (from step 2).

To succeed in the challenge, the students also need to solve three problems in sequence:

1. How should the sensor and robot electronics be integrated to work together?
2. How will the robot manage to immerse the sensors into the liquid to take a measurement?
3. What do the results say about the relationship (if any) between the temperature of a liquid and its pH?

Materials

For each small group (e.g. four students), you will need:

- pH meter and adaptor, e.g. those made by Vernier
- Lego EV3 temperature sensor
- Computer, to write the code for the programmable robots
- Liquids at room temperature: water, plus (optional) vinegar or Coca-Cola
- Large bowl
- Ice cubes or water heater (use both for a larger range of data)
- A robot (lego, scrap, etc.,)

Procedure

The activity consists of five phases.

Phase 1: Introduction

First, discuss the following chemistry topics as a class:

- The theory of acidity and the definition of pH as a measure of the concentration of hydrogen ions in a liquid
- pH meters and the pH scale (the lower the pH, the greater the acidity)
- The challenge: to build a robot that can measure pH and temperature
- The experiment: what data to collect and how to analyse the results

Then divide the students into small groups so they can start thinking critically about their construction plans and the elements needed.

Phase 2: Constructing the robots

This phase is carried out with the students in their groups of four. The students build the basic robots using the Lego EV3 kits, then add the pH meters and temperature sensors to create the Chembots. To make a Chembot, students need to make decisions about:

- Robot morphology – what materials to use, what the robot will look like
- Robot parts – which items from the EV3 kits to use

Phase 3: Programming and testing the robot

Working in groups, the students then programme their robots to carry out the temperature and pH measurements using the programming environment, which is user-friendly and based on blocks. Each block represents a specific function. For example, one of the basic functions is the movement of the gears on the motors to control the sensors.

Once students think they have a working robot, they test it to resolve any remaining problems and challenges. This means ensuring that the sensors are working, the gears on the motors function well, the programming of the robot is correct, the construction of the robot is robust, and so on.

This is the most important and challenging phase – not only for the students but also for the teachers, who need to check that all the students understand what the aim of the project is and to help them with any problems they encounter.

Phase 4: Performance and adjustments

Each group takes a turn using their Chembot to take measurements. You can set up a testing desk in the laboratory with bowls containing the different liquids. To obtain water at different temperatures, put water at about room temperature (25 °C) into a bowl. Cool it gradually by adding ice cubes or heat it using the water heater, repeating the process for each group.

The students then use their Chembots to measure temperature and pH, reading the results from the EV3 screen of the robot and recording them.

Students should analyse their results and try to answer the initial question: what (if any) is the relationship between the temperature of water and its pH? Each group should produce a graph showing their results (this can be done using Excel), as well as their answer to the main question, and any information gathered about the pH of the other liquids.

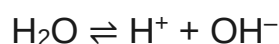
Phase 5: Results and discussion

Finally, the results are discussed as a class. Can the students use their understanding of acids and pH to explain their results?

They should have found that there is a relationship between the temperature of water and its pH: the higher the temperature, the lower the pH. So why is this?

The pH scale represents the concentration of hydrogen (H^+) ions in a liquid. But it is the inverse logarithm (to base 10) of the concentration – so the more hydrogen ions there are, the lower the pH.

For example, at 25 °C, the concentration of H^+ ions in pure water is 10^{-7} , so its inverse log is 7, and thus the pH is 7. As the temperature of water rises, the number of molecules that are dissociated into H^+ and OH^- ions increases.



The dissociation process happens to more molecules at higher temperatures, so there will be a higher number of H^+ ions – and the pH will fall. For example, at around 50 °C, the concentration of H^+ ions is about 10^{-6} (which is ten times larger than 10^{-7}), so its inverse log is 6 and its pH is 6. If the temperature of the water falls, the opposite happens: the concentration of hydrogen ions falls, so the pH increases.

You might like to challenge the students with this question: does the decrease in pH mean that the acidity of water increases with temperature? In fact, this is not the case: for each H^+ ion there is still an OH^- ion, so the water is still neutral. For a liquid to be acidic, it must have an excess of H^+ ions.

End note

At the end of the project, we asked students to complete a short questionnaire about the project. All of our students appreciated this attempt to create an interdisciplinary hands-on activity in the classroom – in fact, they would like this to happen more often in STEM curriculum subjects.

The students also gained an important insight from this project, beyond the details of robotics and chemistry: every field of science is not autonomous, but is connected to other sciences “like links in a chain”, as one student aptly commented.

Credit of this Project: Eleftheria Karagiorgau, Vasileio Spahos.

<https://www.scienceinschool.org/article/2016/chembot-chemistry-robots/>

SAMPLE ASSESSMENT

1. Sample Assessment Methods

- Group Assignment
- Presentation
- Projects
- Written or Practical Quiz
- Written or Practical Test
- Practical Assignment
- Case Studies

2. Sample Assessment Tasks

Module	Content Standard	Topic	Assessment Tasks
12.1 Robotics Meets Chemistry	12.1: Design and build a Chembot that tells whether or not the pH of a liquid as changed with a change in temperature.	1. Acidity Theories (Recall)	Project + Written Test
		2. Robotics (Basics)	
		3. Programming a robot	

3. Sample Assessment Marking Guide

Topic	Assessment Task	Assessment Criteria	Total Marks
Module 12.1 (All Topics)	Project	Build the robot	20
		Program the robot to measure acidity levels	20
Subtotal			40
Module 12.1 (All Topics)	Written Test (Multiple Choices, Short Answers, Calculations, etc)	Correctly answers all long questions	25
		Correctly answers all multiple choice	10
		Correctly answers short answer questions	15
		Subtotal	
Module TOTAL			90

PLANNING, REPORTING & RECORDING (PERFORMANCE CRITERIA)

Assessment Planning

Assessment Domains	Modules	Total Marks	% Weight of Modules
	6		
Theory Knowledge & skills – Test	Test (50)	/50	20
Application of knowledge – Assignments and projects	Assign.1 (30)	/30	30
End of Term Exam	(25 marks)	/25	50
Total marks for term one			100 marks

Assessment Recording

Percentage Range	Achievement Level
85 -100	Very High Achievement (VHA)
65-84	High Achievement (HA)
50-64	Satisfactory Achievement (SA)
25-49	Low Achievement (LA)
0-24	Below Minimum Achievement (BMA)

#	Names	Assessment Task								Total Score (/300)	Percentage Weight (%)				Achievement Level [VHA, HA, SA, LA]
		Tests		Assignments				Project	Exam (/50)		T (20%)	A + P (30%)	Exam (50%)	Ttl CA	
		1	2	1	2	3	4	1							
1	Bill Joe	45	47	20	21	22	23	55	45	278	18.4	25.88	45	89.28	VHA
2	Cara Den	20	25	15	16	17	18	25	30	166	9	18.22	30	57.22	SA
3	Lea May	9	15	5	6	7	8	30	25	105	4.8	9.10	25	38.9	LA

Assessment Reporting

Thema / Biotechnology / Module 12.1: Protein Sequencing Broad Content Standard for Assignments / project 12.1 To determine the amino acid sequence of a protein				
Performance Criteria	Determine Protein Sequence (15 marks)	Prepare Lab Report (5 marks)	Answers all Relative Questions to protein sequencing (5 marks)	25 marks
Very High Achievement	Accurately determine protein sequence (13-15)	Accurately writes a lab report with correct sections (5)	Accurately answers all relative questions to protein sequencing (5)	(21-25)
High Achievement	Partially determine protein sequence (10-12)	Partially writes a lab report with correct sections (4)	Partially answers all relative questions to protein sequencing (4)	(16-20)
Satisfactory Achievement	Fairly determine protein sequence (7-9)	Fairly writes a lab report with correct sections (3)	Impartially answers all relative questions to protein sequencing (3)	(11-15)
Low Achievement	Poorly determine protein sequence (4-6)	Poorly writes a lab report with correct sections (2)	Poorly answers all relative questions to protein sequencing (2)	(6-10)
Below minimum Achievement	Needs improvement in determining protein sequence (0-3)	Needs improvement in writing a lab report with correct sections (1)	Needs improvement in answering all relative questions to protein sequencing (1)	(0-5)

Thema Biotechnology / Module 12.1: Protein Sequencing				
Broad Content Standards For Test: 12.1 To determine the amino acid sequence of a protein				
Performance Criteria	Correctly answers all long questions (25 marks)	Correctly answers all multiple choice (10 marks)	Correctly use concepts to design and propose solutions(15 marks)	50 marks
Very High Achievement	Accurately answers all long questions (21-25)	Accurately answers all multiple choice (9-10)	Accurately use concepts to design and propose solutions (13-15)	(41-50)
High Achievement	Partially answers all long questions (16-20)	Partially answers all multiple choice (7-8)	Partially use concepts to design and propose solutions (10-12)	(31-40)
Satisfactory Achievement	Fairly answers all long questions (11-15)	Fairly answers all multiple choice (5-6)	Fairly use concepts to design and propose solutions (7-9)	(21-30)
Low Achievement	Poorly answers all long questions (6-10)	Poorly answers all multiple choice (3-4)	Poorly use concepts to design and propose solutions (4-6)	(11-20)
Below minimum Achievement	Needs improvement in answering all long questions (0-5)	Needs answers all multiple choice (0-2)	Needs improvement in using concepts to design and propose solutions (0-3)	(1-10)

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